

VrikshaVision: Tree Digital Twin

Project: Export Test Stand | Survey ID: b2f5ae40... | Generated: October 09, 2026 - 00:30 UTC

1. Stand Summary & Calibrated Enumeration

Metric	Value	Statistical Details
Detected Tree Count	29	Direct computer vision instance detections
Calibrated Tree Estimate	29.0	[20.1 - 37.9] (90% prediction interval)
Calibration Status	Uncalibrated Raw (Heuristic Normal)	Ground-truth audit compliance
Total Carbon Stock	197.43 tonnes C	[177.33 - 218.06] tonnes (1,000 draws)
Mean Crown Area	2042.14 m²	Average ground crown projection
Mean Est. DBH	133.15 cm	Allometric scaling from crown projection area
Data Provenance	SYNTHETIC	ML inference model backend

2. Methodology & Scientific Parameterization

Canopy Crown Segmentation: High-resolution aerial imagery is tiled into windowed 512×512 slices with 20% spatial overlap. Overlapping crowns are partitioned via distance-transform marker-controlled watershed, and duplicate cross-tile seams are pruned using Polygon Non-Maximum Suppression (IoU threshold = 0.35).

Split-Conformal Calibration: When ≥10 field audit plots are uploaded, a Negative Binomial Generalized Linear Model with log(detected count) offset is fitted against true ground observations. Residual non-conformity scores on a held-out calibration split define the 90% finite-sample conformal prediction interval.

Monte Carlo Carbon Propagation: Individual tree Crown Projection Area (CPA) is mapped to Diameter at Breast Height (DBH) and Above-Ground Biomass (AGB) using biome-specific pantropical allometric equations (Forest type: *tropical_moist*; Carbon fraction: 0.47). Stand total carbon is evaluated across 1,000 Monte Carlo draws to propagate allometric residual errors.

3. Biophysical Limitations & Regulatory Disclaimers

- Optical aerial surveys cannot distinguish suppressed sub-canopy vegetation obscured by dense dominant crowns.
- Allometric equations rely on regional literature constants and require local destructive harvest or terrestrial LiDAR validation for verified carbon credit issuance.
- Notice on Synthetic Detections:** This report was generated using procedural synthetic mock detections for platform verification and **MUST NOT** be used for regulatory compliance.

SYNTHETIC DATA DEMO

NOT FOR REGULATORY CARBON COMPLIANCE